

Discrete Time Modeling, Implementation and Design of Current Controllers

Christoph H. van der Broeck, Rik W. De Doncker
Institute for Power Electronics and Electrical Drives (ISEA)
RWTH Aachen University
Aachen, Germany
cvb@isea.rwth-aachen.de

Sebastian A. Richter, Jochen von Bloh
AixControl GmbH
Aachen, Germany
info@aixcontrol.de

Abstract — This work addresses discrete time modeling, implementation and design options for the current control of three phase grid tied PWM converters. Based on an accurate discrete time model of the PWM converter, closed loop current control is reviewed from the perspective of the synchronous and stationary reference frame. Then, implementation options for the synchronous frame proportional integral (SFPI) regulator and the proportional resonant (PR) regulator are discussed leading to the formulation of a general controller framework based on space vector resonators: the Resonant Space Vector (RSV) regulator. It can embody multiple SFPI regulators on different frequencies and allows a consistent design of state feedback controllers as well as an efficient implementation. For this control framework an insightful design procedure based on the root locus method for complex system models is introduced. Finally, the performance of the presented control design and implementation concepts is demonstrated experimentally.

I. INTRODUCTION

It is the goal of this work to provide an insightful framework for the design and efficient implementation of discrete time current control structures for grid tied three phase PWM converters. The synchronous frame PI (SFPI) regulator [1] and the proportional resonant (PR) regulator [2] are the most common regulators used for three phase PWM converters, e.g. in active rectifiers or inverters for distributed power generation [3]. In a consistent control structure with proper state feedback decoupling and command feedforward they provide high bandwidth current control with infinite stiffness and zero steady state error at grid frequency.

For PWM converters operating at low switching frequencies discrete time modeling and control design is an important tool. Basic discrete time modeling and control techniques for SFPI regulators and PR regulators have been explored in [4], [5]. They offer high accuracy, since they allow analyzing and designing the dynamics of a computer controlled system in its natural domain. For example the latched characteristics of the voltage pattern generated by a PWM modulated converter and the update delay can be modeled accurately. Furthermore, discrete time models allow designing the regulator based on an accurate converter

model, including the latch and update delay, with closed loop control design methods, e.g. state space control [6] or the root locus method [7]. This is not possible working with continuous time models where typically the open loop transfer function is used to design the controller optimizing the phase margin and the crossover frequency [8], as thereby only approximations of the delays and the latch can be taken into account. This still guarantees the stability of the closed loop system but phase margin cannot be directly related to certain closed loop characteristics e.g. damping, bandwidth or stiffness properties.

With the stronger integration of power electronics in the distribution grid the harmonic content of the grid voltage increases in some grid areas. As a single SFPI and PR regulator cannot entirely reject positive and negative sequence voltage components of higher order, additional control measures have to be applied. For the stiff control of current on multiple harmonics multiple SFPI/PR regulators are used in [9] and [10] or alternatively repetitive control structures are applied in [11] and [12].

To provide a control design and implementation framework in the first part of this paper an accurate discrete time model for grid tied converters is derived. With this model suitable decoupling structures and a trajectory generator is implemented together with a SFPI using classical design and assuming ideal decoupling. To overcome the imprecision of this design approach the root locus method is introduced for the accurate feedback design of complex single input single output (SISO) systems. Formulating the SFPI regulator in the stationary frame [13] as a discrete time space vector resonator allows an optimized tuning and comparison for PR and SFPI regulators. Based on multiple of these space vector resonators operating at different frequencies, a regulator on multiple harmonics can be built in the stationary reference frame without the need for further coordinate transformations leading to the Resonant Space Vector (RSV) regulator. It is demonstrated that the RSV regulator structure can be consistently designed using the root locus method. Finally, the discussed implementation and design concepts are validated in simulation and experiments.

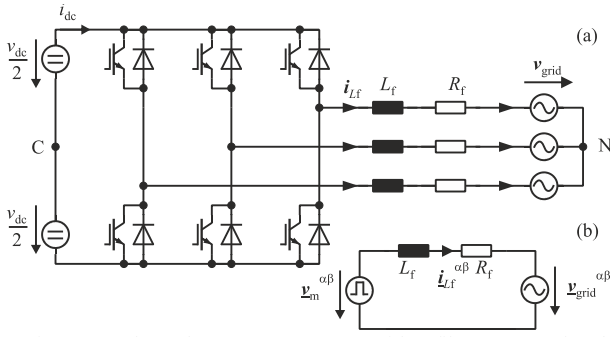


Figure 1. Three phase PWM converter with L -filter connected to the grid (a) Simplified circuit after $\alpha\beta$ -transform (b)

II. MODEL DEVELOPMENT

In this section a discrete time model of a PWM-converter connected via an L -filter to the utility grid (Fig.1a) is developed. The neutral point of the system is isolated. Thus, the three phase currents and voltages can be transformed to two orthogonal space vector components [3] via the $\alpha\beta$ -transform resulting in the simplified equivalent circuit of the three phase converter shown in Fig 1. (b). The $\alpha\beta$ -space vectors can be formulated as complex quantities to reduce the system complexity. Thereby, the converter as scalar multiple input multiple out (MIMO) system is reduced to a complex single input single output (SISO) system like it is presented in [14] and [15]. Most control analysis and synthesis tools in the Laplace- and z -domain can be adopted for these complex systems as it is illustrated in [16] and [17]. Based on the differential equation of the simplified system

$$L_f \frac{d}{dt} \underline{i}_{Lf}^{\alpha\beta} = \underline{v}_m^{\alpha\beta} - R_f \underline{i}_{Lf}^{\alpha\beta} - \underline{v}_{load}^{\alpha\beta}, \quad (1)$$

which describes the dynamics of the current $\underline{i}_{Lf}^{\alpha\beta}$ in the inductance L_f and the resistance R_f as a function of the modulated converter voltage $\underline{v}_m^{\alpha\beta}$ and the grid voltage $\underline{v}_{grid}^{\alpha\beta}$, the transfer function $\underline{G}_{filter}(s)^{\alpha\beta}$ (2) of the L -filter can be determined as a function of $\tau_f = L_f/R_f$.

$$\underline{G}_{filter}(s)^{\alpha\beta} = \frac{\underline{L}_f(s)^{\alpha\beta}}{\underline{V}_m(s)^{\alpha\beta} - \underline{V}_{grid}(s)^{\alpha\beta}} = \frac{1}{R_f} \frac{1}{s\tau_f + 1} \quad (2)$$

For the voltage modulation a classical asymmetrical regular sampled PWM [18] is used, which allows applying an arbitrary three-phase average voltage to the filter: The voltage modulation process for one half bridge of the converter is depicted in Fig. 2. At the top and bottom of the carrier signal $s_{carrier}$ the duty cycle d_{PWM} is updated and the converter current i_{Lf} is measured and sampled. This synchronous sampling allows capturing the average current. The computer control system can be operated at a sampling frequency of twice the PWM frequency. However, as the duty cycle is only updated twice per PWM interval, there is a constant update delay of $T_{update} = T_s$.

A. Discrete Time Converter Model

The discrete time model of the PWM converter connected via the L -filter to the utility grid is developed based on the state block diagram of the plant depicted in

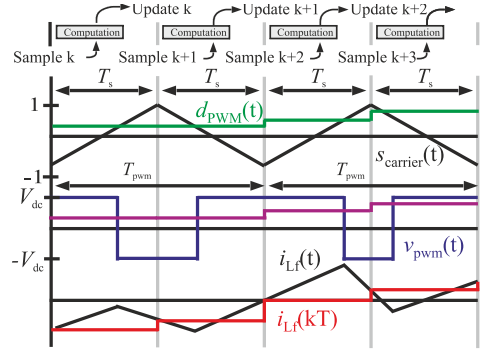


Figure 2. Pulse width modulation of one converter half bridge

Fig. 3. It takes into account the filter dynamics (2), the update delay of the controller $\underline{G}_{update}(z) = z^{-1}$, and the modulator dynamics. The low-pass characteristic of the inductive load ensures that only the average voltage contributes to the average rate of change of the current. Therefore, the applied pulses are approximated as a latch in the stationary frame [5]. The discrete time transfer function of the converter modeled as a latch and interfaced to an inductive filter can be determined according to

$$\underline{G}_{conv}(z)^{\alpha\beta} = \frac{\underline{L}_f(z)^{\alpha\beta}}{\underline{V}_m(z)^{\alpha\beta}} = (1 - z^{-1}) Z \left\{ \frac{\underline{G}_{filter}(s)^{\alpha\beta}}{s} \right\}. \quad (3)$$

$$\underline{G}_{conv}(z)^{\alpha\beta} = \frac{1}{R_f} \frac{(1 - e^{-T_s/\tau_f}) z^{-1}}{1 - e^{-T_s/\tau_f} z^{-1}} \stackrel{R_f \rightarrow 0}{\approx} \frac{1}{L_f} \frac{T_s z^{-1}}{1 - z^{-1}} \quad (4)$$

For the case of a negligible resistance the discrete time model can be simplified to (4). The discrete time model of the plant is concluded adding the update delay $\underline{G}_{update}(z)$:

$$\underline{G}_p(z)^{\alpha\beta} = \frac{\underline{L}_f(z)^{\alpha\beta}}{\underline{V}_m^*(z)^{\alpha\beta}} = \frac{1}{L_f} \frac{T_s z^{-1}}{1 - z^{-1}} z^{-1}. \quad (5)$$

For most common control designs the plant is transformed into the synchronous reference frame rotating with the grid frequency ω_e to design a PI regulator for dc-quantities. This model transformation can be done back shifting the coordinate transformation (6) and applying the z -transformation on the difference equation (7) resulting in (8).

$$\underline{x}^{\alpha\beta}(k) = \underline{x}^{dq}(k) e^{j\omega_e k T_s} \quad (6)$$

$$\underline{x}^{\alpha\beta}(k-1) = \underline{x}^{dq}(k-1) e^{j\omega_e k T_s} e^{-j\omega_e T_s}. \quad (7)$$

$$z^{-1}|_{\alpha\beta} \rightarrow z^{-1} e^{-j\omega_e T_s}|_{dq} \quad (8)$$

The transfer function of the system in the synchronous frame (9) exhibits complex coefficients due to the crosscoupling

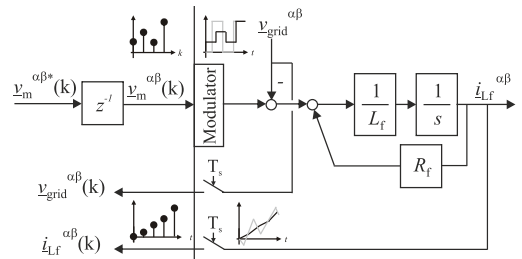
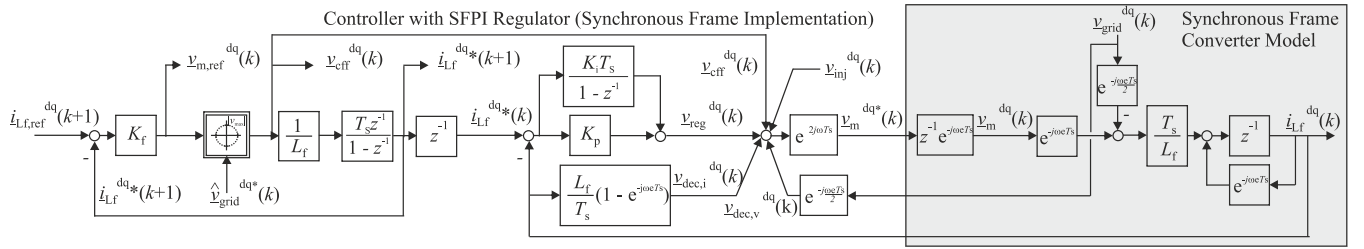


Figure 3. State block diagram of the converter connected to the grid



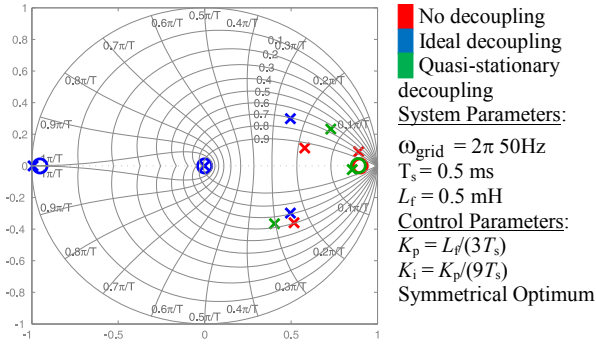


Figure 5. SFPI control design with the symmetrical optimum ($a = 2$)

disturbance rejection as disturbances are compensated without overshoot. However, the more realistic scenario with quasi-stationary decoupling exhibits dynamics, which deviate from this ideal case: The poles and zeros close to $z = 1$ do not cancel each other and the dominating pole pair is slightly shifted and not complex conjugated anymore. This phenomenon of complex poles without conjugated counterpart occurs as a consequence of the complex SISO system. If the decoupling is omitted the resulting poles and zeros are even more shifted with respect to the desired pole placement. The impact of the decoupling can be even better observed in the time domain from the command response and disturbance response of the closed loop system depicted in Fig. 6. The time domain system responses nicely show the impact of the quasi-stationary decoupling, which nearly prevents the d-component to cross couple in the q-component. Nevertheless, a discrete time control design with an accurate model, which reflects the remaining cross coupling, allows a more accurate design of the closed loop system dynamics than the quasi continuous time design using the SO. Therefore, in the next section a suitable discrete time control design approach is illustrated, which allows a more accurate control design.

C. Trajectory Generation and Command Feedforward

For the optimization of the command tracking performance a consistent command trajectory filter is built. It contains the system model, which is fed back by a gain that manipulates the bandwidth of the generated current reference and command feedforward voltage. Furthermore, the command trajectory is constraint to the available voltage margin to prevent hitting the voltage limit of the system. It is very important to delay the current reference by one

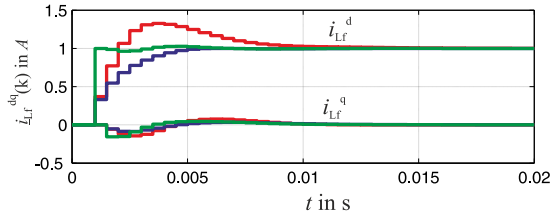


Figure 7. Command tracking with different command filter bandwidths (System and Control Parameters from Fig. 5 $K_f = L_f f_{TF}$)

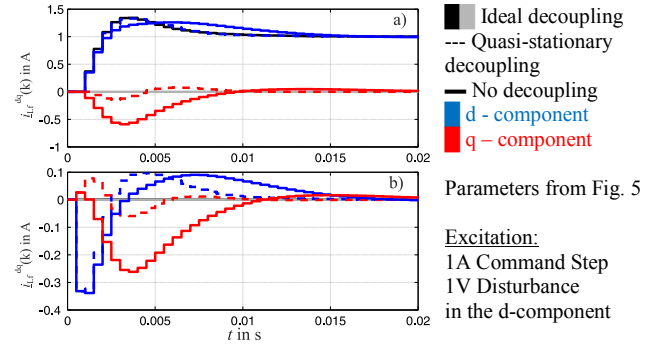


Figure 6. Command response (a) and disturbance response (b)

sampling step as shown in Fig. 4, to ensure that the applied command feedforward voltage can source the desired current before the corresponding current reference is set. Due to the command feedforward, the controller just needs to compensate disturbances and model inaccuracies. Thus, it can be optimized for disturbance rejection. The state block diagram depicted in Fig. 4 shows the entire control structure with state feedback decoupling, SFPI regulator and trajectory generator, which is used as a basis for the following considerations.

IV. DISCRETE TIME REGULATOR DESIGN

To accurately design the dynamics of discrete time control loops, a regulator design procedure based on the root locus method is introduced, which takes into account the cross coupling of the complex SISO model. First, the implementation and design of PR and SFPI regulators is discussed. Then the concept is generalized to the resonant space vector (RSV) regulator.

A. Implementation Options for Current Regulators

The SFPI regulator (19) is usually designed and implemented in the synchronous reference frame

$$\underline{G}_{\text{SFPI}}(z)^{\text{dq}} = K_p + \frac{K_i T_s}{1 - z^{-1}} = K_p \left(1 + \frac{K_i T_s}{K_p} \frac{1}{1 - z^{-1}} \right). \quad (19)$$

However, the design and the implementation of the regulator can be done in any other reference frame achieving identical system dynamics: For example in [13], the continuous SFPI regulator has been implemented in the stationary reference frame. Similarly, the synchronous frame discrete time control structure presented in Fig. 4 can be transformed into the stationary reference frame using the inverse transformation of (8). The entire control structure from Fig. 4 implemented in the stationary frame is shown in Fig. 8. For the SFPI regulator the transformation leads to

$$\underline{G}_{\text{SFPI}}(z)^{\alpha\beta} = K_p + \frac{K_i T_s}{1 - e^{j\omega_e T_s} z^{-1}}, \quad (20)$$

which can be implemented on a DSP as complex difference equation using the backshift operator of the z-transform:

$$\underline{v}_{\text{SFPI,reg}}^{\alpha\beta}(k) = K_p \Delta \underline{i}_{\text{Lf}}^{\alpha\beta}(k) + \underline{v}_{\text{res}}^{\alpha\beta}(k) \quad (21)$$

$$\underline{v}_{\text{res}}^{\alpha\beta}(k) = K_i T_s \Delta \underline{i}_{\text{Lf}}^{\alpha\beta}(k) + \underline{v}_{\text{res}}^{\alpha\beta}(k-1) e^{j\omega_e T_s}$$

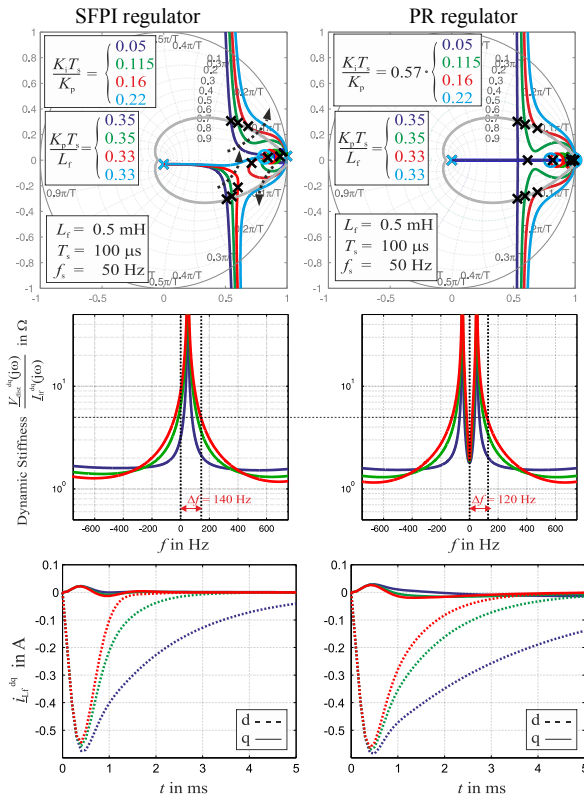


Figure 9. Control design of the complex SISO system based on (a) Root locus plot of the complex SISO system, (b) double sided dynamic stiffness plot (c) disturbance response to 1V step in the d-axis

that the closed loop system poles change for variations of the sampling frequency. Therefore, in Fig. 10 the root locus plot is shown for a sampling frequency of 2 kHz and 8 kHz using identical tuning gains of PR and SFPI regulator. It can be seen that at high sampling frequencies the PR regulator tuning gain needs to be chosen to half the tuning gain of the SFPI regulator to keep a similar damping. However, at low sampling frequencies both regulators can be designed with nearly identical gains. Thus, working with low PWM frequencies, e.g. in high power applications, it might be beneficial to use the PR regulator, whereas at high sampling frequencies one would only use the PR regulator if the regulation of the negative sequence current is mandatory.

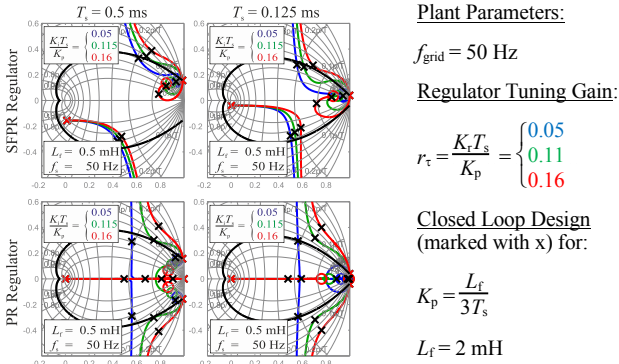


Figure 10. Comparison of SFPI and PR regulator at fast and slow sampling

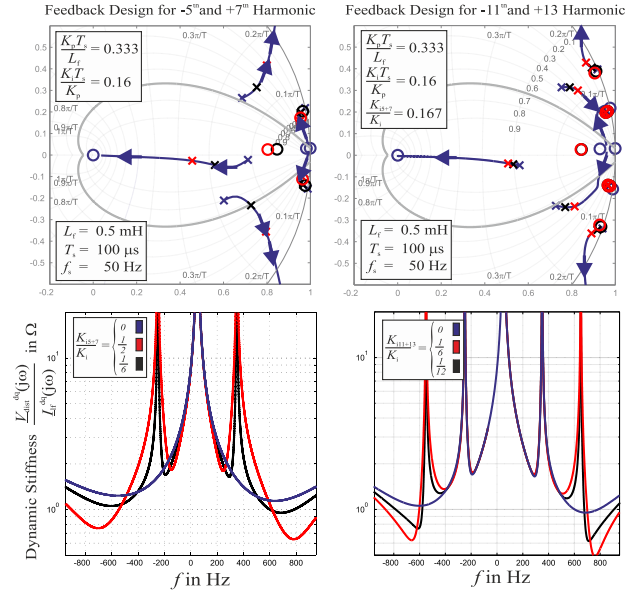


Figure 11. Design of the RSV regulator with fundamental -5th, +7th, -11th and 13th harmonic based on the root locus method

C. Design and Implementation of the RSV Regulator

The introduced RSV regulator is a compact way to formulate a regulator on multiple harmonics. One drawback of these higher order regulator structures is that it is not intuitive to design multiple state feedback gains. To overcome this problem, it is shown here how the root locus method for complex SISO systems can be used to optimize the state feedback gains of multiple space vector resonators. A properly tuned SFPI regulator is used as a starting point which is augmented by space vector resonators achieving infinite stiffness at -5th, +7th, -11th and +13th grid harmonic. The design goal for the entire regulator is to augment the stiffness around desired frequencies with high state feedback gains keeping the effective damping of the system high. First, the RSV state feedback gains rejecting the 5th and 7th grid voltage harmonic are synthesized, which are kept identical ($K_{r,6} = K_{i,+5} = K_{i,-7}$). The root locus in Fig. 10 shows the possible pole placements for an increasing $K_{r,6}$. For the red and black marked poles two design options are shown in detail, whose dynamic stiffness plot is shown below in comparison to the dynamic stiffness of the pure SFPI regulator used as starting point. The poles resulting from the space vector resonators on the -5th and +7th harmonic are balanced by zeros and are not critical for the design, as they do not effectively weaken the system damping. The same is true for the critically damped pole. However, by higher $K_{r,6}$ gains the poles resulting from the proportional state feedback path are further pulled to weaker damping. Thus, the design decision for $K_{r,6}$ is a compromise between high system damping and a bigger bandwidth of the resonant state feedback loop. This is illustrated for two cases: $K_{r,6}/K_i = 1/2$ and $1/6$. As $K_{r,6}/K_i = 1/2$ reduces the damping significantly resulting in a reduction of the worst case stiffness by a factor of 2, a design with $K_{r,6}/K_i = 1/6$ is a reasonable option.

TABLE I. PARAMETERS OF THE EXPERIMENTAL SETUP I

$L_f = 2 \text{ mH}$	$L_{\text{grid}} = 0.2 \text{ mH}$	$f_{\text{pwm}} = 1 \text{ kHz}$	$T_s = 0.5 \text{ ms}$
$V_{\text{dc}} = 300 \text{ V}$	$V_{\text{grid},l-n} = 130 \text{ V}$	$K_p = 1.33 \Omega$	$K_i = 426 \Omega/\text{s}$

After the successful resonant state feedback loop design of the -5^{th} and $+7^{\text{th}}$ harmonic, the resulting closed loop system dynamics are used as starting point to design the -11^{th} and $+13^{\text{th}}$ harmonic resonant feedback loop. The root locus for an increasing $K_{r,12} = K_{i,-11} = K_{i,+13}$ depicted in Fig. 10 shows that this time the complex conjugated pole pair corresponding to the proportional state feedback loop is not leading to weak damping. The critically damped pole and also the poles resulting from the space vector resonator on 5^{th} and 7^{th} harmonic do not limit the feedback gain. However, the pole resulting from the added space vector resonators on the -11^{th} and 13^{th} harmonic reveal a low system damping for higher values of $K_{r,12}$. This effect is a result from the growing distance and weaker balancing between pole and zero. Thus, a design with $K_{r,6}/K_i = 1/6$, marked in red, is not desirable as it results in a very weakly damped pole closed to the unity circuit, whereas state feedback gains of $K_{r,12}/K_i = 1/12$ seem to be a good design compromise. This exemplary design shows how a RSV regulator structure is designed systematically and consistently generalizing the root locus method for complex SISO system.

V. SIMULATION AND EXPERIMENTAL RESULTS

Based on the introduced control framework a current control structure according to Fig. 8 has been designed and implemented for the PWM converters, whose parameters are summarized in Tab. 1-3. It is tested in simulation using PLECS and experimentally in a rapid control prototyping environment based on the XCS2000 system by AixControl. For the modulation of the converter voltage a space vector modulation with equal zero states according to [20] is used.

A. Evaluation of the Command Tracking Performance

First, the performance of the SFPI regulator and the command filter is investigated for the system parameters summarized in Tab. I. The simulation in Fig. 12 shows the system command response. The trajectory generator, which

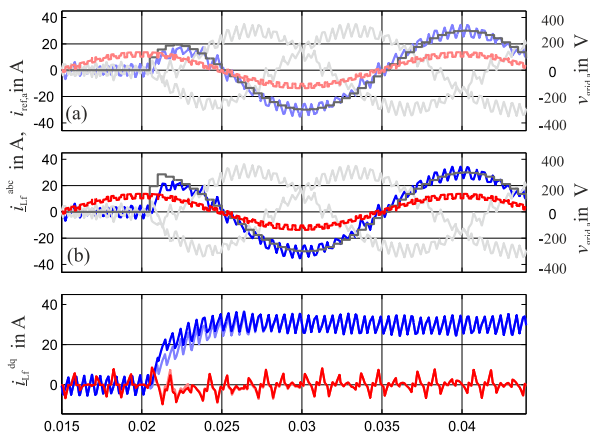
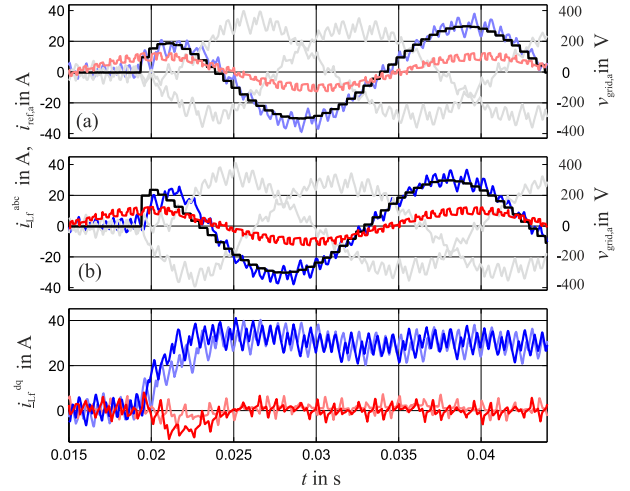
Figure 12. Simulation of the command response with a trajectory generator gain $K_f = Lf/(3Ts)$ (a) and $K_f = Lf/Ts$ (b)

TABLE II. PARAMETERS OF THE EXPERIMENTAL SETUP II

$L_f = 0.5 \text{ mH}$	$R_{\text{chopper}} = 1 \Omega$	$f_{\text{pwm}} = 5 \text{ kHz}$	$T_s = 0.1 \text{ ms}$
$V_{\text{dc}} = 250 \text{ V}$	$f_{\text{ref}} = 50 \text{ Hz}$	$K_p = 1.7 \Omega$	K_i variable

Figure 13. Experimental results of the command response with a trajectory generator gain $K_f = Lf/(3Ts)$ (a) and $K_f = Lf/Ts$ (b)

includes the voltage limit, shapes a feasible current command and applies a consistent command feedforward leading to a high command tracking bandwidth without overshoot. With the gain of the trajectory generator K_f the bandwidth of the system can be properly adjusted independent of the regulator gains. The simulation results can nearly perfectly be transferred to the experiment, which is shown in Fig. 12. The major difference in current between simulation and experiment occurs due to a small grid voltage imbalance and minor pulsations of the dc-link. This shows that an accurate simulation of a grid tied PWM converter reaches nearly the precision of an experimental setup.

B. Comparison of SFPI and PR Regulator

It has been discussed in Section IV B that at high sampling frequencies the PR regulator needs to be detuned compared to the SFPI regulator to keep a desired damping

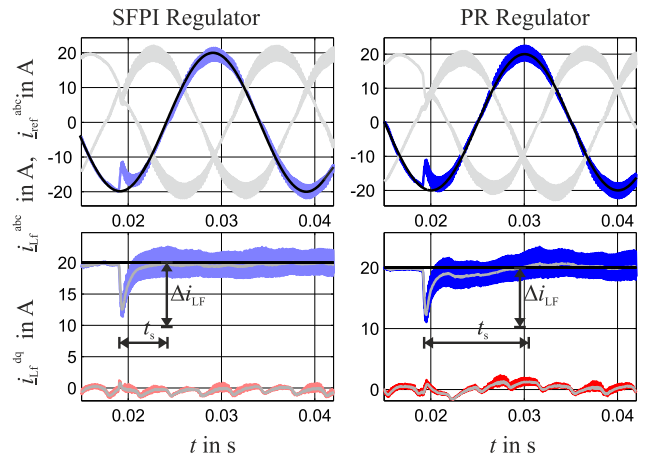
Figure 14. Experimental response to a chopped resistance of 1Ω SFPI ($K_i = 2500 \Omega/\text{s}$) and PR regulator ($K_r = 1400 \Omega/\text{s}$)

TABLE III. PARAMETERS OF THE EXPERIMENTAL SETUP III

$L_f = 2 \text{ mH}$	$L_{\text{grid}} = 0.2 \text{ mH}$	$f_{\text{pwm}} = 4 \text{ kHz}$	$T_s = 250 \text{ } \mu\text{s}$
$V_{\text{dc}} = 420 \text{ V}$	$V_{\text{grid, l-n}} = 200 \text{ V}$	$K_n = 2.6 \text{ } \Omega$	$K_i = 1750 \text{ } \Omega/\text{s}$

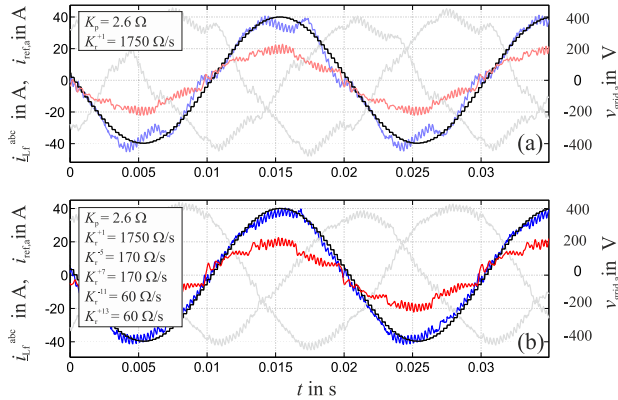


Figure 15. Experimental evaluation of a SFPI regulator (a) and a RSV regulator on mutiple harmonics (b) under distorted grid conditions

level. Consequently the PR regulator has worse disturbance rejection properties. In Fig. 14 the response of the controlled converter (Tab. II) to a three phase chopped resistance is presented. The PR regulators needs more than twice the settling time t_s compared to the SFPI regulator. As the proportional state feedback gain of both is identical the current sag Δi_{Lf} in both setups are identical.

C. Mutiple Harmonic Rejection with the RSV regulator

Finally, the RSV regulator concept is evaluated. Therefore, the converter (Tab. III) is tied to a grid with significant voltage harmonics. In Fig. 15 (a) only the SFPI regulator is activated. It only regulates the fundamental current harmonic in phase and amplitude accurately, but cannot compensate the high voltage harmonics. Therefore, in (b) a RSV regulator on the -5^{th} , 7^{th} , -11^{th} and 13^{th} harmonic is added. The design of the feedback gains has been done according to the outlined design method. The properly tuned regulator leads to the regulation of a smooth current waveform.

VI. CONCLUSION

An accurate discrete time model for the control of a three phase converter has been derived. This has been used to develop proper decoupling paths and a consistent command filter structure. The root locus method has been generalized for the control design of complex SISO systems and used to optimize the SFPI and the PR regulators for three phase current control. Thereby, both regulator structures have been compared systematically. Based on the SFPI implementation in the stationary frame, the resonant space vector regulators concept has been developed. It allows to handle various regulators e.g. SFPI, PR and regulators on multiple harmonic in one framework leading to consistent tuning and an efficient implementation. Experimental results are presented to validate the introduced control implementation and design framework.

REFERENCES

- [1] D. Schauder and R. Caddy, "Current control of voltage-source inverters for fast four-quadrant drive performance," IEEE Trans. Ind. Appl., vol. IA-18, pp. 163–171, Mar./Apr. 1982.
- [2] D. N. Zmood, D. G. Holmes, G. H. Bode, "Frequency-domain analysis of three-phase linear current regulators," IEEE Trans. Ind. Appl., vol. 37, no. 2, pp. 601–610, Mar./Apr. 2001.
- [3] V. Blasko, V. Kaura, "A new mathematical model and control of a three-phase AC-DC voltage source converter," IEEE Trans. Power Electronics, vol. 12, no. 1, pp. 116–123, Jan. 1997.
- [4] S. A. Richter and R. W. De Doncker, "Digital proportional-resonant (PR) control with anti-windup applied to a voltage-source inverter," in Proc. 14th Eur. Conf. Power Electron. Appl., pp. 1–10, 2011.
- [5] K. Hongrae, M. W. Degner, J. M. Guerrero, F. Briz, F. R. D. Lorenz, "Discrete-Time Current Regulator Design for AC Machine Drives," IEEE Trans. Ind. Appl., vol. 46, no. 4, pp. 1425, 1435, Jul.-Aug. 2010.
- [6] S. A. Richter, J. von Bloh, C. P. Dick, D. Hirschmann, R. W. De Doncker, "Control of a medium-voltage test generator," IEEE Power Electronics Specialists Conference 2008, pp. 3787, 3793, 2008.
- [7] C. H. van der Broeck, R. W. De Doncker, S. A. Richter, J. von Bloh, "Unified control of a buck converter for wide load range applications," IEEE APEC 2014, pp. 2788–2795, March 2014.
- [8] J. Shen, S. Schröder, H. Stagge, R. W. De Doncker, "Precise Modeling and Analysis of DQ-Frame Current Controller for High Power Converters with Low Pulse Ratio," IEEE Energy Conversion Congress and Exposition, September 2012, Raleigh, NC, USA.
- [9] P. Mattavelli, "A closed-loop selective harmonic compensation for active filters," IEEE Trans. Ind. Appl., vol. 37, no. 1, pp. 81, 89, Jan./Feb. 2001.
- [10] M. Hausmann, N. Grass, B. Piepenbreier, "Harmonic compensation in a load emulation system using different control techniques," IEEE 34th International Telecommunications Energy Conference INTELEC, pp. 1, 7, Sept. 30 Oct. 4 2012.
- [11] K. Zhou, D. Wang, "Digital repetitive learning controller for three-phase CVCF PWM inverter," IEEE Trans. Ind. Electron., vol. 48, no. 4, pp. 820, 830, Aug. 2001.
- [12] S. Engel, K. Righers, R. W. De Doncker, "Digital repetitive control of a three-phase flat-top-modulated grid tie solar inverter," 13th European Conference on Power Electronics and Appl., EPE '09, pp. 1, 10, 8–10 Sept. 2009.
- [13] T. M. Rowan, R. J. Kerkman, "A New Synchronous Current Regulator and an Analysis of Current-Regulated PWM Inverters," IEEE Trans. Ind. Appl., vol. IA-22, no. 4, pp. 678, 690, July 1986.
- [14] F. B. del Blanco, M. W. Degner, R. D. Lorenz, "Dynamic analysis of current regulators for AC motors using complex vectors," IEEE Trans. Ind. Appl., vol. 35, no. 6, pp. 1424, 1432, Nov./Dec. 1999.
- [15] F. Briz, M. W. Degner, R. D. Lorenz, "Analysis and design of current regulators using complex vectors," IEEE Trans. Ind. Appl., vol. 36, no. 3, pp. 817, 825, May/Jun. 2000.
- [16] S. Gataric, N. R. Garrigan, "Modeling and Design of Three-phase Systems Using Complex Transfer Functions," Proc. IEEE Power Electron. Spec. Conf., vol. 2, June/July 1999, pp. 691–697.
- [17] L. Harnefors, "Modeling of Three-Phase Dynamic Systems Using Complex Transfer Functions and Transfer Matrices," IEEE Trans. Ind. Electron., vol. 54, no. 4, Aug. 2007, pp. 2239–2248.
- [18] Holmes, G. D., Lipo, T. A. (2003), "Pulse Width Modulation for Power Converters - Principles and Practice," IEEE Press Series on Power Engineering, John Wiley and Sons, Piscataway, NJ, USA.
- [19] V. Blasko, V. Kaura, "A novel control to actively damp resonance in input LC filter of a three phase voltage source converter," IEEE APEC 1996, vol. 2, pp. 545, 551, 3–7 Mar. 1996.
- [20] H. van der Broeck, H. Skudelny, G. Stanke, "Analysis and Realization of a Pulse Width Modulator Based on Voltage Space Vectors," IEEE Trans. Ind. Appl., vol. 24, no. 1, pp. 142–150, Jan./Febr. 1988.